Cover Letter

To: *Digital Finance* Editorial Board **Re:** Resubmission against the open Major-Revision decision — *Do Cryptocurrency Markets Differentiate Infrastructure from Regulatory Shocks? A Multi-Moment Event Study with Dependence-Robust Inference* (formerly *Infrastructure vs Regulatory Shocks: Asymmetric Volatility Response in Cryptocurrency Markets*, Manuscript ID DAFI-D-25-…) **Author:** Murad Farzulla (Dissensus) **Date:** [pending submission]

Dear Editors,

Please find attached my revised manuscript, submitted against the open Major-Revision slot for the infrastructure-versus-regulatory paper. I owe you a frank account up front, because this revision is larger than the decision letter asked for, and the headline conclusion of the paper has changed.

What happened, plainly. Reviewer 2's central comment (R2.1) asked whether the 5.7× infrastructure-regulatory volatility multiplier survives when the event-selection screen is relaxed. (That 5.7× was the prior paper's high-salience-curated estimate under its canonical estimator and rolling winsorisation; the present standalone pipeline does not reproduce it and instead gives a reproducible 4.88× point estimate — global-clip winsorisation, canonical estimator — which is the consistent headline figure I now carry forward. The 5.24×/5.7× estimator bookkeeping is set out in the response.) Addressing that question properly required me to reconstruct the candidate pool and re-run the inference end to end. Doing so surfaced two defects in the inference behind the original headline — not in the data, not in the point estimate, but in how the significance was computed. Once corrected, the headline asymmetry result does not survive. The asymmetry is directional but not statistically distinguishable from zero under dependence-robust inference. **This revision is that reckoning.** I am overturning my own published estimate, and I am doing so because the correct inference required it.

I want to be equally clear that this is not a retreat into a thinner paper. The reckoning *is* the contribution. What began as a selection-bias robustness check became a worked, self-refuting demonstration of how cross-asset event studies in heavy-tailed markets manufacture significance — and how the correct inference dissolves it. I believe that demonstration is more useful to *Digital Finance*'s readership than the original claim would have been, precisely because the failure mode is common and the corrections are specific and reproducible.

The two inference defects, disclosed. The original significance rested on two moves that, examined closely, manufacture precision:

1. **Pseudoreplication.** The headline test (t = 4.768, p = 0.0008) treated the six per-asset GJR-GARCH-X coefficients as six independent observations. They are not: all six assets see the *same* 50 events and are strongly cross-correlated (mean $\bar{\rho} \approx 0.69$), so the effective sample size is far below six. Their coefficients cluster tightly because they average the same correlated events, not because the effect is precisely estimated. Re-testing at the event level — the unit of independence — collapses the significance (event-level Welch p = 0.52; cluster-robust panel p = 0.50); a design-effect correction on the actual estimator lands the honest value at $p \approx 0.07$.
2. **Gaussian bootstrap innovations under heavy tails.** The cross-asset bootstrap meant to handle dependence drew its standardised innovations from a Gaussian even though every model is fitted with Student-t errors at $\nu \approx 3$. With $\nu \approx 3$ the true innovation is far heavier-tailed; the Gaussian draw mis-specifies the null distribution and biases the bootstrap p downward. Correcting the draw to the fitted Student-t margins (a multivariate-t copula) moves the baseline p from 0.057 (the heavy-tail-naive Gaussian value) to **0.322**.

Together these are the difference between $p \approx 0.001$ and a failure to reject. The point estimate ($\approx 5\times$) never changed; only the significance did, and only because the original significance was an artefact.

To keep this from reading as a single dataset talked from a low p to a high one, I added two pieces of apparatus. An **inference ladder** walks the *identical* comparison through seven progressively-more-correct inference procedures — the rungs *bracket* the comparison rather than climbing monotonically (the naive test far below at 0.0008, the low-power model-free proxies above near 0.5, the calibrated t-copula bootstrap settling it at 0.322 as each defect is corrected). And a **Monte-Carlo size study** simulates a thousand panels under a fitted true null of no differential effect and measures how often each procedure falsely rejects: the naive i.i.d. rule rejects a true null **51% of the time at the nominal 5% level** ($\approx 10\times$ nominal), while the Student-t-copula bootstrap I adopt restores nominal size (3.9%). The bootstrap is self-validating — calibrated from the same null DGP, its size *must* land near nominal, and it does — so the over-rejection of the naive rule is a measured property, not a quirk of these data. The headline $p = 0.0008$ was, with high probability, one of the 51% of true nulls the naive rule falsely rejects.

The dual null, reported honestly. Having corrected the inference, I do the obvious thing the original framing avoided: I test the same events at *both* moments of the return distribution, on one shared sample, under one matched dependence-robust design.

- **First moment (returns).** The infrastructure-regulatory cumulative-abnormal-return difference is +7.19 percentage points (infrastructure *less* negative), with an event-level block-bootstrap $p = 0.283$ and an Ibragimov–Müller few-cluster $p = 0.295$; the 95% CI $[-5.97, +20.48]$ crosses zero comfortably. This subsumes the companion returns analysis (the *no-structure* paper) into the present multi-moment paper — the two were always the same events viewed through different moments, and they belong in one place.
- **Second moment (variance).** The point estimate is directional (≈ 2.6–$5\times$, infrastructure larger) but the Student-t-copula CCC-GARCH-X bootstrap returns $p = 0.322$ at baseline and $p = 0.318$ with a crisis-regime control.

Neither moment supports a statistically distinguishable asymmetry. The original story — "the structure lives in the second moment" — does not hold: the variance signal is, on correct inference, no more distinguishable from zero than the return-level difference.

What I am explicitly *not* claiming. I am not claiming any significant asymmetry at either moment. I am also not claiming the effects are zero or that the two shock types are identical. The first moment is plainly **underpowered** — the minimum detectable effect is $\approx$ 19–20 pp against an observed $\approx$ 7 pp — so this is an honest *failure to reject*, not a demonstration of equality. Throughout the manuscript the dual null is stated symmetrically and with this caveat; I carry no 5%-significance claim anywhere.

What survives, and supports the paper. Two findings remain and serve as ballast for the inference contribution:

- **The scope condition.** Replacing expert event identification with a purely mechanical impact screen of a broad reconstructed candidate pool (135 candidates) attenuates the variance multiplier to a modest 1.3–1.6$\times$ that is not significant at any threshold (the like-for-like two-asset screen gives 1.61$\times$, $p = 0.16$). The strong magnitude is a property of curated, high-salience identification, not of a mechanical threshold. This is exactly Reviewer 2's R2.1 concern, answered by *measuring* selection bias as an object rather than asserting it away.
- **The weekly Granger result.** At the GDELT series' native weekly frequency, sentiment Granger-leads volatility for 10 of 18 asset-sentiment pairs (7 surviving Benjamini–Hochberg at $q < 0.05$). A prior daily test that forward-filled the weekly series found nothing — an artefact of the step-function predictor the forward-fill creates (R2.3).

A pre-submission audit, and two new controls. Because the paper's whole pitch is inferential discipline, I did not want a referee to be the first to find a hole. Before resubmitting I ran the manuscript and its code through an independent external audit; the point-by-point response now closes with a dedicated section addressing its nine findings (F1–F9). Two prompted new analyses, both of which strengthen the paper: a direct **event-in-mean control** (demeaning on the event dummies so events enter both moments — the first-moment effects are small and symmetric, -0.45% vs -0.46%, and the variance multiplier moves only $4.88\times \rightarrow 4.42\times$, confirming the asymmetry is second-moment, not a relabelled return shock), and **concrete screen-vs-curation evidence** on selection (of the 50 curated events, 14 fail the mechanical impact screen yet were retained while 57 screen-passers were dropped — direct evidence the curation tracked event type and salience, not realised volatility). The remaining findings were caveats or fixes. The manuscript is 41 pages and compiles clean, the abstract is trimmed to lead on the methodological contribution, the funding statement reads "No external funding was received," and the full pipeline now runs end-to-end from a clean clone of the public repository (`github.com/studiofarzulla/crypto-event-study`).

On scope and venue. The decision letter offered a Major Revision on a paper whose headline I am now retracting on my own initiative. I have thought carefully about whether this still belongs at *Digital Finance*. I believe it does, more than the original did: the question (do crypto markets differentiate shock types?) is unchanged and squarely in scope; the answer is now an honest, dependence-robust dual null with a transferable methodological lesson at its core; and the paper now subsumes a companion returns study into a single multi-moment treatment, which is a cleaner and more complete object than either half alone. I would rather submit the result the data actually support than defend a headline correct inference does not sustain. If the editors judge that the changed conclusion warrants fresh review, I fully understand; I have written the manuscript and this letter to make that judgment easy.

The point-by-point response that follows addresses the original reviewer comments (R1.x, R2.x) where they remain relevant, and is candid about where the reframing has superseded or absorbed them. I am grateful to both reviewers — the selection-bias comment in particular is what set this whole correction in motion, and the paper is far more honest for it.

Sincerely,

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